Distances and approximate portage access

Controls: Distance to Shores and Waterways; Waterway Reference Year; Approximate Portage Access.

1. Four distances and four squared terms. The distance box adds distance to the ocean, Great Lakes, operating rivers and operating canals. Each distance runs from the county's or MSA's geometric centre to the nearest mapped feature. The centre depends on the area's shape, without population or employment weights. Distances are approximations computed in the project's map projection.

For each feature, let r be distance in kilometres divided by 100. The regression includes both r and r^2, with contribution

$$\gamma_1 r + \gamma_2 r^2.$$

Thus 200 kilometres gives $r = 2$ and $r^2 = 4$. Zero distance remains zero. With the square included, the distance relationship can curve: the slope at distance r is $\gamma_1 + 2\gamma_2 r$. The linear coefficient alone is therefore not a constant effect of moving another 100 kilometres away. Selecting this box adds eight variables in total.

2. Which waterways and years? River distance uses documented steamboat routes, and canal distance uses operating canals in the selected waterway year. Ocean and Great Lakes distances use modern shorelines. A distance remains missing if a route with uncertain operating dates could be nearer than the nearest route known to be active. Missing selected distances remove observations. These controls describe the contiguous United States.

3. What the portage proxy measures. A portage is a place where goods must be carried over land between navigable stretches. The fall line is the transition between the Piedmont and Coastal Plain, where river gradients and rapids could obstruct navigation. Following the geographic idea in Bleakley and Lin (2012), the project identifies candidate locations where mapped rivers cross that boundary.

When a river does not cross the mapped boundary, the construction can retain the nearest non-tidal endpoint if it lies within 10 kilometres of the boundary. Endpoints connected to tidal river segments are excluded from this additional rule. The construction uses the complete river map, independently of the selected waterway year.

4. How to interpret the indicator. Approximate Portage Access equals one when any part of a county or MSA lies within 10 kilometres of a candidate, and zero otherwise. It adds one indicator, without a squared term. The candidates come from map geometry and have not been verified against historical portage records. The indicator therefore measures proximity to plausible locations; it does not establish that a portage or port existed there.